Impact of knowledge search on product and process innovation: mediating role of absorptive capacity and moderating role of IT capability

Received 3 December 2019
Revised 24 April 2020
30 June 2020
25 September 2020
7 December 2020
Accepted 9 December 2020

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Abstract

Purpose – More enterprises adopt open innovation by breaking technological or organizational boundaries to seek internal and external knowledge when they face a fiercely competitive environment, complex market demands, and increasingly rapid technological change. In this context, a knowledge search strategy is regarded as an effective means of obtaining inside and outside resources and an important way to break the innovation bottleneck. Moreover, information technology (IT) is deemed an important asset for sourcing knowledge, whereas absorptive capacity is seen as an indispensable ability for utilizing novel knowledge. Thus, this paper aims to test the role of knowledge search in open innovation and examine the mediating effect of absorptive capacity and the moderating effect of IT capability.

Design/methodology/approach – Using a sample of 1,088 Chinese firms' data collected by the World Bank in 2012, this paper employs logistic regression to test the hypotheses.

Findings – This study finds that local and boundary-spanning search strategies positively influence both product and process innovation, and absorptive capacity has a mediating role in the relationships between knowledge search and product and process innovation. Moreover, IT capability has a positive moderating effect on the relationship between local search and innovation performance; however, IT capability strengthens the relationship between boundary-spanning search and process innovation while weakens that between boundary-spanning search and product innovation.

Originality/value – This study explores the impact of different knowledge search behaviors on different types of innovation and probes the role of absorptive capacity and IT capability in mediating and moderating the above relationships. By drawing on knowledge-based theory and cognitive-developmental theory, this paper provides a novel perspective to explain the mechanism between knowledge search and innovation performance.

Keywords Local search, Boundary-spanning search, Absorptive capacity, IT capability, Product innovation, Process innovation

Paper type Research paper

Introduction

Organizations today find it difficult to innovate alone amidst the hostility of competitive contexts, the complexity of markets, mobility of knowledge workers, and the velocity of technological changes. Organizational learning theory argues that an open learning structure can help firms deal with environmental uncertainties and solve problems such as "path dependence" and "capability traps" (Mina *et al.*, 2014). Because the open innovation model

This study was supported by the National Science Foundation Committee of China (Grant No. 71572063).

European Journal of Innovation Management
Vol. 25 No. 2, 2022
pp. 325-346

1460-1060
DOI 10.1108/EJIM-12-2019-0350

highlights the insight of leveraging external knowledge and expertise across firm boundaries (Chesbrough and Bogers, 2014; Chesbrough *et al.*, 2006), organizations nowadays widely accept that innovation can be generated independently or through making use of external knowledge. Hence, the search for novel knowledge both inside and outside and the integration of internal and external expertise and resources is vital to innovation performance. Local and boundary-spanning search for knowledge is the key way to improve innovation performance.

Product and process innovation are considered as two different types of innovation characterized by fundamentally different attributes and determinants (Gomez *et al.*, 2016; Köhler *et al.*, 2012; Reichstein and Salter, 2006). Product innovation was not the only type of innovation that could benefit from the knowledge search (Radicic, 2020). Process innovation would also be introduced by discovering new applications of existing technologies to decreasing costs or enhancing quality and flexibility (Wu, 2014; Aliasghar *et al.*, 2019). Yet, the related literature has concentrated mainly on product innovation, and the research on how firms' knowledge search can develop process innovation remains limited (Tsinopoulos *et al.*, 2017).

Moreover, in the process from searching to innovating, valuable knowledge is recognized as one of the most important assets for coping with environmental and technological change, achieving sustainable competitive advantage, and improving innovation performance such as new products and processes (Liebeskind, 1996; Ndofor and Levitas, 2004). However, knowledge itself does not automatically translate into performance but needs the facilitation of enterprise competence in the process of identification, acquisition, transformation, and exploitation. Absorptive capacity (AC) (Grimpe and Sofka, 2009) and information technology (IT) capability (Dong and Netten, 2017) are two crucial factors that play important roles in promoting search efficiency and enhancing innovative performance. Although previous studies have made significant efforts to demonstrate the mechanism of AC on the relationship between knowledge search strategy and innovation performance (Ferreras-Méndez *et al.*, 2016, 2019; Flor *et al.*, 2018; Mothe *et al.*, 2018; Wang and Guo, 2020), little attention has been paid to the role of IT capability and the interplay between different capacities (Cui *et al.*, 2015; Ahn *et al.*, 2016).

For addressing the above gaps, this paper first constructs a theoretical model containing knowledge search, AC, IT capability, and product and process innovation based on organizational learning theory. Next, we conduct an empirical analysis with a large-scale sample of Chinese manufacturing firms to explore the relationships between different search patterns and innovative types and the underlying mechanisms. Specifically, we deem that knowledge search positively affects the innovation performance, in which AC plays a mediating role, and IT capability plays a moderating role. Moreover, this article also examines the interaction of ability factors and further explores how such relationships are different for two types of innovation.

The remainder of this article is organized as follows. We review the theoretical background and develop our hypotheses in the next section. Then, the research design and the results of the empirical analysis are presented in sequence, and the discussions are given at the end.

Literature review and hypotheses development

Product and process innovation

Innovation is a knowledge-based activity with a complex process in which enterprises reconstruct, combine, and create new things from existing elements. As an important organizational behavior for enterprises to obtain innovation resources in the open innovation model, knowledge search enriches the knowledge base for innovation activities and becomes the logical origin for knowledge absorption and internalization in the organizational learning process. Hence, in the perspective of organizational learning, the knowledge stock of the firm is accumulated through learning in the established path of the organization. The acquisition of this knowledge stock comes not only from the input of the internal R&D department of the organization but also from the innovative ideas of other departments and the heterogeneous resources outside the organization. Companies can collect, learn, and integrate within and outside the organization by formulating and implementing different search strategies.

Product and process innovation are two main dimensions to measure the innovation performance because of the fundamentally different need of knowledge structure and processing capabilities (Gomez *et al.*, 2016). On the one hand, product innovation is the creation of new technology that can provide consumers with new products or services. On the other hand, process innovation is the improvement of the existing production process. In general, product innovation refers to the company's successful introduction of new products, while process innovation is defined as the improvement of the processing technologies required to produce a product.

Local and boundary-spanning knowledge search

The existing literature has discussed knowledge search strategy from different perspectives to illustrate its advantages for increasing the knowledge base, adding to the patent pool, and enhancing firm performance (Cohen *et al.*, 2002). From a problem-solving perspective, organizations are looking for solutions inside and outside the organizational boundary, local or distant. Previous studies show that firms predominantly search locally in the vicinity of their current knowledge base (Levinthal and March, 1981; Tripsas and Gavetti, 2000) and then are forced to search distantly to find external solutions and potential nonlocal providers if the problem cannot be solved using existing routines (Lopez-Vega *et al.*, 2016). Local search increases the probability of finding workable solutions, whereas distant search improves the novelty of innovations (Fleming and Sorenson, 2004). Therefore, knowledge search can vary in terms of search boundary dimensions (technological or organizational boundaries) and can be classified into local search and boundary-spanning search (Rosenkopf and Nerkar, 2001). Typically, a search is considered local when it accesses knowledge from nearby established trajectories or the current knowledge base, whereas a boundary-spanning search is an approach to find new and unfamiliar knowledge.

This mainly focuses on the organizational boundary dimension to divide knowledge search strategy into local search and boundary-spanning search. Herein, local search is defined more specifically as searching knowledge from intraorganizational function areas, whereas boundary-spanning search, by contrast, means searching for knowledge from various external stakeholders, including suppliers, customers, competitors, universities, consultants, public sectors, and other institutions (Laursen and Salter, 2006; Segarra-Ciprés and Bou-Llusar, 2018).

Knowledge search and product and process innovation

The uncertain market information and rapid technological changes that firms face devalue their existing knowledge base quickly (Wu and Wu, 2014) and continually drives them to practice complex search processes (Xie *et al.*, 2018). Given the constant pressure for survival, literature in the innovation field has increasingly stressed the importance of acquiring and synthesizing heterogeneous knowledge across boundaries in product or service development. Meanwhile, acquiring knowledge internally (e.g. internal R&D) is important, while external knowledge search becomes a crucial approach for innovation.

The primary task of innovation is to search for heterogeneous knowledge or fresh ideas. No matter where the new knowledge comes from (internal or external), firms use fresh knowledge as complementarities of existing technological (Vega-Jurado *et al.*, 2008), that is, innovation depends to some extent on the knowledge resources acquired by knowledge search. Knowledge brought by different search strategies has different effects on product and process innovation.

Local search for fresh internal knowledge can find solutions to problems in a cheap and efficient manner. This kind of search activity can provide different degrees of knowledge supports for product and process innovation. For product innovation, the positive role of local search can be reflected by influencing internal R&D. Reusing similar knowledge elements can not only reduce superfluous steps in the product development process but also deepen the understanding of new knowledge combinations. The experience gained by the learning process of local search can improve the efficiency of internal R&D from small growth to a quantum leap. Meanwhile, process innovation is the improvement of product or service provision (e.g. cost reduction and quality enhancement). The improvements in processes are less tangible and often hard for the customers to notice. Local search finds solutions in past experiences and practices, which contribute to process improvement. In general, local search has a positive impact on product and process innovation to some extent.

Boundary-spanning search can provide companies with complementary and heterogeneous knowledge (Hume and Hume, 2015), which can not only help organizations improve existing products or creates new products but also provide more methods to solve process problems. On the one hand, the impact of a boundary-spanning search on product innovation mainly derives from external heterogeneous knowledge with high quality and quantity. The diverse knowledge can help organizations break existing patterns, form new frameworks, explore new territories, and achieve continuous breakthroughs in products or services. The diversity of knowledge base can prevent organizations from falling into the "familiarity trap", thereby increasing the possibility of improving or creating products. On the other hand, the impact of a boundary-spanning search on process innovation is mainly reflected in the information provided by stakeholders. For example, knowledge gained from suppliers can contribute to product improvement. Boundary-spanning search provides creative sources for process innovation clearly and efficiently. Therefore, boundary-spanning search behavior has a positive impact on product and process innovation.

To sum up, two types of search strategies have positive effects on both product and process innovation. These arguments lead us to the following hypotheses:

H1. Local search has a positive impact on product and process innovation.

H2. Boundary-spanning search has a positive impact on product and process innovation.

Knowledge search, absorptive capacity, and innovation performance

It has been long known that a firm's strategy can affect its ability, and in turn, influence innovation performance. Nothing comes from nothing, and hence firms need to invest large number of resources in search of heterogeneous knowledge to promote innovation (Savino *et al.*, 2017). From the perspective of organizational learning theory, AC is an important capability mainly determined by prior related and accumulated knowledge and maybe be created as a byproduct of a firm's R&D investment in learning and problem-solving (Cohen and Levinthal, 1990). AC enables a firm to recognize, assimilate, and integrate novel knowledge at a lower cost and higher efficiency (Flor *et al.*, 2018). Learning reflects the ability to assimilate knowledge for imitation, whereas problem-solving represents a capacity to create new knowledge for innovation (Kim, 1998). Therefore, AC has been long recognized as one of the fundamental learning processes and a pivotal driving factor for enterprises to efficiently integrate external knowledge flows and prior knowledge base (Ferreras-Méndez *et al.*, 2016; Ritala and Hurmelinna-Laukkanen, 2013).

Some studies have suggested an association between searching and the level of AC. For example, Ferreras-Méndez *et al.* (2015) show the positive effect of external knowledge search on absorptive capacity in terms of search depth and breadth. Knowledge search exposes firms to various sources of ideas that broaden their knowledge base. On the one hand, local search inside the organization and technology trajectory can deepen the firm's cognitive ability for understanding tacit knowledge. On the other hand, boundary-spanning searches across organizational boundaries and industry segments can find pervasive, diverse knowledge that improves the capability of identifying potentially valuable knowledge. Hence, a firm's knowledge search strategy will be positively related to AC:

H3a. Local search is positively related to AC.

H3b. Boundary-spanning search is positively related to AC.

Openness to external sources allows firms to draw in ideas from outsiders to deepen the knowledge pool of available technological opportunities, and thus, can enhance their innovation results (Laursen, 2012). However, from the perspective of organizational learning, it is imperative to develop the capabilities to transform and exploit rather than merely acquire novel knowledge (Ferreras-Méndez *et al.*, 2019). AC consists of the processes of exploratory, transformative, and exploitative learning that facilitate the development of products and processes (Lane *et al.*, 2006).

First, it helps organizations track changes in their own technology fields more effectively with knowledge-seeking capacities. Firms with well-developed acquisition and assimilation capabilities are less likely to become locked into a narrow area of expertise (Flor *et al.*, 2018). Second, AC enables the timely application of new knowledge to innovation activities with transformational and exploitative capabilities. Specifically, transformation integrates acquired novel knowledge into the existing knowledge base, whereas exploitation converts these fresh ideas into new products and processes. Moreover, searching internally and externally for novel knowledge is never costless (Cruz-Gonzalez *et al.*, 2015; Flor *et al.*, 2018). The high level of AC can reduce the costs of openness.

Previous studies have supported the positive effect of AC on product and process innovation. For example, it is suggested that demand-pull and science-push AC are positively related to product and process innovation output (Murovec and Prodan, 2009). Because the previous knowledge base is essential to recognize external knowledge (Ramayah *et al.*, 2020), new ideas and fresh knowledge can be effectively transformed and exploited through a high level of AC. Because the innovation activities are based on heterogeneous knowledge and capacity reconfiguration, a higher level of AC may be related to a higher degree of product and process innovation. Thus, we propose the following hypotheses:

H4. AC will exert a positive impact on product and process innovation.

Furthermore, we presume that knowledge search has not only a direct influence but also an indirect effect on product and process innovation. That is,

H5a. AC mediates the positive impact of local search on product and process innovation.

H5b. AC mediates the positive impact of boundary-spanning search on product and process innovation.

Knowledge search, IT capability, and innovation performance

Obtaining technical knowledge from other departments and outside enterprises can achieve the improvement of innovation performance. However, only when companies have sufficient basic R&D, absorptive or technical capabilities can they make full use of these learning opportunities. With the rapid development of information technology, IT is an important strategic resource input for firms' knowledge search and innovation activity because of its efficiency for collecting, processing, and sharing organizational knowledge (Alavi and Leidner, 2001; Dong and Netten, 2017; Joshi *et al.*, 2010; Kleis *et al.*, 2012). However, an IT resource itself cannot directly foster innovation unless combined with other resources.

IT capability is identified as a firm's ability to control IT-related costs and influence organizational goals by implementing IT assets. Since the introduction of resource-based view and enterprise capability theory into the field of information management, scholars have further enriched the implications of IT capability. It encompasses IT infrastructure, human IT resources, and IT-enabled intangibles that can be leveraged to enhance efficiency, reduce costs, or increase revenues (Chae *et al.*, 2018).

Drawing on the study by Bharadwaj (2000), we argue that IT capability refers to an enterprise's ability to capture value through matching IT assets with other resources. More specifically, IT capability can be divided into internal and external IT capability in terms of the activities that IT capability supports. Internal IT capability includes related resources used to support business activities inside firms such as product/service enhancement and operations, whereas external IT capability highlights the ability to use IT resources to support activities linking firms to external sources such as promoting market development or maintaining external partner relationships.

IT capability is likely to play a crucial role in adopting knowledge search strategies and enhancing innovation performance (Gómez *et al.*, 2017; West and Gallagher, 2006) with the development of information systems. In the context of open innovation, IT capability can be specifically used to establish knowledge management systems, connect with intraorganizational and interorganizational knowledge sources, and to improve the efficiency of internal and external knowledge utilization (Wu *et al.*, 2019; Wade and Hulland, 2004).

IT capability is the basis of information processing and can strengthen the relationships between knowledge search (local and boundary-spanning search) and firm innovation (product and process innovation). Specifically, IT capability provides efficient search paths and methods for accessing diverse heterogeneous knowledge, facilitates knowledge sharing and communication between departments or with outside organizations, and supports the internalization and reuse of heterogeneous knowledge (Roberts *et al.*, 2012; Trantopoulos *et al.*, 2017; von Krogh, 2012). For example, the development and widespread applications of IT have alleviated the problems caused by oversearch (Kang and Kang, 2009). In the knowledge-based view, IT plays an important role in shaping an enterprise's knowledge base. Because internal IT capability and local search are mainly in the boundary of the firm while external IT capability and boundary-spanning search are mainly out the boundary of the firm, we propose the following hypotheses:

H6a. Internal IT capability positively moderates the effect of local search on product innovation.

H6b. External IT capability positively moderates the effect of boundary-spanning search on product innovation.

H7a. Internal IT capability positively moderates the effect of local search on process innovation.

H7b. External IT capability positively moderates the effect of boundary-spanning search on process innovation.

We propose the research model in Figure 1 to demonstrate the relationships between knowledge search and firm innovation and the roles of AC and IT capability.

Method

Data and sample

To test our hypotheses, we use data from the official survey on the innovative activities of the People's Republic of China manufacturing firms conducted by the office of the World Bank. Aim to obtain 3,000 interviews (150 interviews within each sector), a total of 20,616 establishments were enumerated. Following a two-stage procedure, a screening phase over the phone and a face-to-face interview with the top manager of each establishment, the data is collected between December 2011 and February 2013. In the screening stage, the percentage of noneligible units was confirmed (6,485, account for 31%) and then excluded. Finally, the survey successfully conducted onsite interviews of 2,700 privately owned and 148 state-owned sample firms selected using stratified random sampling and include comprehensive information various aspects, with a survey response rate of 20.15% (2,848/14,131).

The questionnaire was prepared in English over 41 pages long. Moreover, it is implemented by high-performing local agency teams who conduct face-to-face interviews with top managers to generate valid information and high-quality data. It systematically distinguishes between product and process innovation when asking for the ways of introducing innovative activities. Further, it also asks for the extent of information technologies used to support different business activities. Considering the research purpose of examining the mechanism between knowledge search strategy and innovation performance, we filtered the data in terms of variable characteristics. After removing observations with missing variables, the useful sample size is 1,088, covering 19 industries and 25 cities, and can be considered as representative of the Chinese manufacturing industry. Table 1 reports the sample's descriptive information. Further, nonresponse bias was tested by comparing the industries represented in the sample with the population sample in terms of size and sales, and we found no significant differences between the two groups.

Additionally, although the data are relatively old, we deem that the theory findings maybe not totally depend on the age of the data. In the study by Bogers and Lhuillery (2018), for example, they used the survey data from 1993 to test their research hypotheses. Although the data are old, the results are original. This can also be exemplified by many other new studies that have used an old version of CIS/KIS survey data sets (Kang and Kang, 2014; Amoroso, 2017; D'Este *et al.*, 2016; Giannopoulou *et al.*, 2019).

Measurement

Dependent variables – product and process innovation. Based on the characteristics of different innovation activities, we follow previous studies and divide firm-level innovation activities into product and process innovation (Gomez *et al.*, 2016; Bogers and Lhuillery, 2018). The two dependent variables capture whether the firm has generated product and process

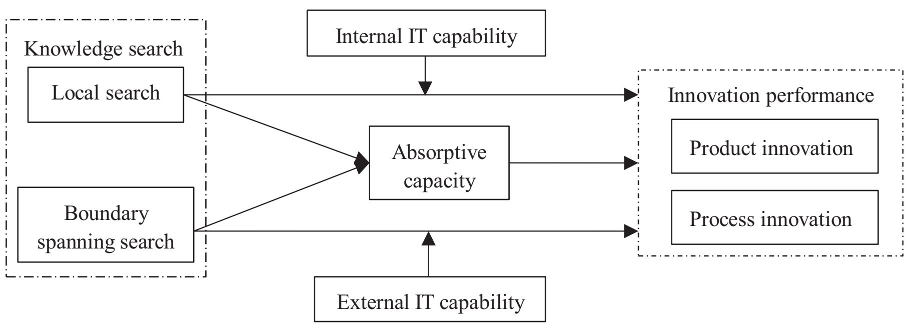

Figure 1.
The research framework

Industry	Competitors <=50	>50	Firm size 5–19	20–99	>=100	Firm age <5	5–10	>10	Total
Metallic mineral products	22	84	12	45	49	6	33	67	106
Basic metals	4	56	9	28	23	0	19	41	60
Chemicals	15	84	10	47	42	0	19	80	99
Electronics	17	77	10	29	55	0	35	59	94
Fabricated metal products	18	94	15	41	56	2	35	75	112
Food	21	95	12	50	54	3	25	88	116
Furniture	1	7	1	3	4	0	4	4	8
Garments	4	69	9	29	35	0	21	52	73
Leather	1	8	1	6	2	0	1	8	9
Machinery and equipment	17	74	11	41	39	1	43	47	91
Paper	3	8	1	9	1	0	5	6	11
Plastics and rubber	15	86	10	47	44	1	25	75	101
Precision instruments	4	6	2	8	0	0	5	5	10
Recorded media	3	9	2	8	2	0	4	8	12
Recycling	1	4	1	2	2	0	5	0	5
Refined petroleum product	2	3	1	2	2	0	0	5	5
Textiles	6	93	9	42	48	2	30	67	99
Transport machines	21	54	7	24	44	1	33	41	75
Wood	0	2	0	1	1	0	1	1	2
Total	175	913	123	463	502	16	344	728	1,088

Table 1.
Sample profile

innovations. Specifically, *product innovation (PROD)* is measured as a binary variable that takes the value 1 when the firm has introduced any new products or services in the last three years, and 0 otherwise. Meanwhile, *process innovation (PROC)* is measured through a binary variable that takes the value 1 when the firm has introduced new technology and equipment for improving the production process over the last three years, and 0 otherwise. These two types of innovation are not an either-or condition; firms can practice product and process innovative activities at the same time.

Independent variables. Knowledge search. In the questionnaire, the item "in what ways has this establishment introduced new products or services and new or improved process?" is used for identifying knowledge search. This paper divides the behavior of knowledge search into local search (*LS*) and boundary-spanning search (*BSS*) based on the degree of spatial specificity of knowledge. *LS* is a binary variable 1 if the establishment developed or adapted in house, 0 otherwise. Moreover, *BSS* is defined as an index constructed as the sum of ways that firms introduced innovative activities. The index ranges from 0 when no ways are implemented, up to 4, when all ways are used. The knowledge search variables are separately measured for product and process innovation, and Table 2 shows the proportion of each search channel. Specifically speaking, a firm simultaneously uses the local and boundary-spanning search for one innovative type or one search strategy for product and process innovation. For example, 73.35% of companies apply the local search for product innovation in our dataset, and this percentage for process innovation is 72.89.

Absorptive capacity. Similar to other studies, we deem the firm's internal R&D expenses as a proxy of *AC* (Cohen and Levinthal, 1990; Denicolai *et al.*, 2016). In this paper, *AC* is defined as a binary variable. It is set to 1 if the firm spends on research and development activities within the organization or contracted with other companies in the last three years, and 0 otherwise. Furthermore, as a test of the robustness of the results for the measure of absorptive capacity, we calculate an alternative measure (AC_{sub}) through the percentage of the firm's annually average R&D expenses (within or contracted with other companies) over the last three years scaled by total sales. The disadvantage of this substitution is that the item response rate of R&D investment is low; only 37.96% (413/1,088) of establishments respond to the expenditure of R&D activities. Fortunately, the results of this subset have proved the robustness to some extent.

IT capability. Advancements in information technology have made organizational boundaries so porous and help organizations acquire higher efficiency when leveraging knowledge for innovation. The variable of IT capacity mainly focuses on the application of IT resources in the organization and comes from the question "to what extent are information and communication technologies used to support key business activities". To calculate the orientation of IT capability, we ran a factor analysis based on the five items of supported business activities. The KMO (0.83) and the Chi-square for Bartlett's test of sphericity (3943.83) were highly significant ($p < 0.0001$). Based on the above results, two factors were retained that reflect 85.19% of the variance in the original data according to the cumulative proportion of variance. The factor analysis explained the application of IT on partner relations, marketing and sales, and customer relations as the external IT capacity (*EITC*), while it attributed the utilization of IT on product and service enhancement and production and operations to internal IT capacity (*IITC*).

Control variables. For analysis, we use market concentration, industry, firm size, and firm age as control variables. Specifically, the degree of market concentration (*MARCON*) is a binary variable measured by the number of competitors, that is, 1 if the competitors are higher than 50, and 0 otherwise. According to the OECD technology intensity definition, we define the *industry* as a binary variable. We deem high and medium-high industries as high-tech and others as low-tech industries. The mean value of the industry variable donates the proportion of high-tech companies in our dataset, the same as other binary variables. Moreover, firm size is presented by the log value of the number of employees, and firm age is defined as the log value of the duration since the enterprise began operations. Table 3 provides a summary of the measures.

Analysis and results

Descriptive analysis

Before testing the hypotheses, we reported the descriptive statistics and correlation matrix, as well as a variance inflation factor (VIF) of explanatory variables in Table 4. The correlation coefficients and VIF values are often used to check the possibilities of multicollinearity that might make the estimation results inaccurate. From Table 4, most of the correlations are significant, and the correlation coefficient between any two variables is less than 0.36. The initial statistics show that multicollinearity is unlikely to be a crucial problem in general.

Knowledge search	Channels of available knowledge sources	Used percentage Product innovation	Process innovation
Local search	Developed or adapted in house	73.35	72.89
Boundary-spanning search	Developed in cooperation with suppliers	27.02	32.54
	Developed in cooperation with client firms	34.65	32.54
	Introduced or licensed from another firm	41.91	34.56
	Implemented idea from external sources	27.67	32.35

Table 2. Difference of knowledge search between innovation types ($N = 1,088$)

Var.	Description	Mean	S.D.	Min	Max
PROD	Whether a firm introduced any new products or services in the last three years	0.52	0.50	0	1
PROC	Whether a firm introduced new technology and equipment for improving production process in the last three years	0.69	0.46	0	1
LS	Whether developed or adapted in house	0.73	0.44	0	1
BSS	The sum of the ways involved supplier, client, competitor, and other external source	1.31	1.32	0	4
AC	Whether invested in R&D activities over the last three years	0.48	0.50	0	1
IITC	The average extent to support product/service enhancement and production/operations	3.10	1.22	1	5
EITC	The average extent to support partner relations, marketing/sales, and customer relations	3.37	1.23	1	5
MARCON	Whether the number of competitors >50	0.84	0.37	0	1
Industry	Whether a firm belongs to high-tech industries	0.35	0.48	0	1
Firm size	The natural logarithm of the number of employees	1.93	0.55	0.70	4.48
Firm age	The natural logarithm of the duration of operation	1.10	0.20	0	2.10

Table 3. Summary of the variables

Meanwhile, the result of the variance inflation factor (VIF) analysis also provides evidence for the multicollinearity problem with the largest value of VIF (2.46). It is generally accepted that there is no multicollinearity problem when the VIF value is less than 10 (Kang and Kang, 2014).

The influence of knowledge search on product and process innovation

Because most of the primary indicators in our research are given as discrete variables, we use the Logistic regression model step by step to test hypotheses, and to observe the effect of control variables on other variables and the variation trend of the variable coefficient. Following a hierarchical procedure, we first put all control variables into a Logistic model as a benchmark and then test the effect of independent variables on dependent variables, respectively. Table 5 presents the results of the main effect regression.

The testing results of Model 1 and Model 5 show the effectiveness of control variables on innovation. In the benchmark model, there are significant positive effects of firm size and negative impacts of market concentration on product and process innovation. Specifically, large enterprises have more input of innovation resources than small enterprises so that the firm size positively affects the innovation, which demonstrates one of Schumpeter's hypothesis. For the external environment, if the market tends to be fully competitive, the products tend to be homogenized. Hence, an enterprise will lack motivation for innovation in a fiercely competitive market.

Moreover, if a company uses a certain type of knowledge search only, whether local or boundary-spanning, it will have a positive impact on innovation performance. Specifically speaking, in Model 2 and Model 6, the regression coefficients of local search are both significant and positive ($\beta = 0.703, p < 0.01; \beta = 0.829, p < 0.01$), confirming its positive effect on product and process innovation. This outcome supports H1. Similarly, the results of Model 3 and Model 7 show the positive and significant impact of boundary-spanning search on product and process innovation ($\beta = 0.401, p < 0.01; \beta = 0.654, p < 0.01$), which supports H2. Besides, it is worth noting that local search has a greater impact on process innovation than that on product innovation. The same applies to the boundary-spanning search. To test the significance of this difference, we use the Wald test method. The results show that the difference of boundary-spanning search ($\chi^2 = 10.06, p < 0.01$) is significant while the local search is not. Hence, knowledge from boundary-spanning search can provide more foundation for process innovation.

Table 4. Descriptive statistics, correlations, and VIF of explanatory variables

Variable	1	2	3	4	5	6	7	8	9
1. *LS*	1								
2. *BSS*	0.228***	1							
3. *AC*	0.304***	0.197***	1						
4. *IITC*	0.075**	0.355***	0.278***	1					
5. *EITC*	0.021	0.328***	0.347***	0.747***	1				
6. *MARCON*	−0.055*	−0.088***	−0.174***	−0.059*	−0.042	1			
7. Industry	0.042	0.064**	0.110***	0.083***	0.034	−0.082***	1		
8. Firm size	0.166***	0.073**	0.207***	0.189***	0.173***	−0.047	0.027	1	
9. Firm age	0.040	0.019	0.038	0.067**	0.083***	0.064**	−0.023	0.219***	1
VIF	1.19	1.22	1.32	2.37	2.46	1.05	1.02	1.14	1.06

Note(s): *$p < 0.10$; **$p < 0.05$; ***$p < 0.01$

Table 5.
Results for testing of the direct effect

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
	PROD				*PROC*			
MARCON	−0.482***	−0.468***	−0.412**	−0.384**	−0.878***	−0.864***	−0.918***	−0.899***
Industry	0.203	0.183	0.151	0.144	0.011	0.001	−0.069	−0.085
Firm size	0.561***	0.443***	0.493***	0.470***	0.465***	0.348***	0.361***	0.322**
Firm age	−0.012	−0.092	−0.096	−0.012	0.303	0.341	0.367	−0.002
LS		0.703***		0.492***		0.829***		0.567***
BSS			0.401***	0.368***			0.654***	0.631***
No. of obs.	1,088	1,088	1,088	1,088	1,088	1,085	1,088	1,085
Pseudo R^2	0.024	0.039	0.067	0.076	0.026	0.051	0.115	0.126
Log likelihood	−735.25	−724.19	−703.08	−696.36	−653.99	−636.54	−594.86	−586.30
χ^2 (*f*)	36.47***(4)	58.59***(5)	100.8***(5)	114.24***(6)	35.49***(4)	68.2***(5)	153.78***(5)	168.67***(6)

Note(s): $^{**}p < 0.05$; $^{***}p < 0.01$

Furthermore, when considering two types of search behaviors simultaneously, the results of Model 4 and Model 8 indicate that both local search and boundary-spanning search have a significant positive impact on product innovation ($\beta = 0.492, p < 0.01; \beta = 0.368, p < 0.01$) and process innovation ($\beta = 0.567, p < 0.01; \beta = 0.631, p < 0.01$), which proved H1 and H2 again. These two models also test the differences in the impact of knowledge search on product and process innovation, respectively. The coefficients of Model 4 show that local search has a stronger positive influence on product innovation than boundary-spanning search. Meanwhile, the results of Model 8 manifest that boundary-spanning search has a stronger effect on process innovation than that of local search.

However, the unobserved heterogeneity problem, residual variation, for example, may cause a false appearance when comparing logit model coefficients directly. Hence, we use the KHB decomposition method to compare coefficient differences to verify the significance of this difference between the two types of knowledge search. It turns out that the decomposed coefficients of local search and boundary-spanning search in Model 4 are 0.594 ($z = 4.05$, $p < 0.01$) and 0.386 ($z = 7.41$, $p < 0.01$) respectively, which proved the significantly stronger positive influence of local search on product innovation. In the same way, the decomposed coefficients in Model 8 are 0.644 ($z = 9.03$, $p < 0.01$) and 0.665 ($z = 4.40$, $p < 0.01$), which verify the significantly tinny stronger positive influence of boundary-spanning search on process innovation.

The mediating role of AC

The step by step method is used to test the mediating effect of absorptive capacity. Its premise is that significant relationships exist among dependent, independent, and mediating variables. The test results are shown in Table 6, providing evidence of the premise. According to previous research viewpoints (Baron and Kenny, 1986), the evidence of mediating effect is that the influence of independent variables on dependent variables will decrease or cease to be significant when mediators enter the regression model. Furthermore, it plays a partial mediation role if the effect decreases versus an absolute mediation role if the effect becomes insignificant.

The results of Model 9–11 prove that knowledge search strategy can positively impact the firm's AC, whether it adopts local search ($\beta = 1.413$, $p < 0.01$), boundary-spanning search ($\beta = 0.275, p < 0.01$), or both ($\beta = 0.063, p < 0.01; \beta = 0.059, p < 0.01$). Hence, H3 is supported. From the perspective of control variables, we find that highly intensive technology industries and the scale of the firm are positively related to AC whereas fierce market competition is negatively related to AC. In addition, Model 12 and Model 14 respectively certify the significant role of AC to enhance the product and process innovative performance ($\beta = 1.955$, $p < 0.01$; $\beta = 1.436$, $p < 0.01$), supporting H4.

For product innovation, the proof of the mediating effect is revealed by Model 4 and Model 13. Boundary-spanning search ($\beta = 1.879$, $p < 0.01$) and absorptive capacity ($\beta = 0.338, p < 0.01$) of Model 13 have significant effects on product innovation while the impact of local search is insignificant. There is a similar result for process innovation (see Model 8 and Model 15). Therefore, absorptive capacity has a mediating effect between knowledge search and innovation performance, which supports H5a and H5b. Specifically speaking, absorptive capacity plays an absolute mediation role between local search and product/process innovation and a partial mediation role between boundary-spanning search and innovation performance.

Additionally, we also conducted a Bootstrap test to improve the validity of the mediating effect of AC. The Bootstrap method is based on whether the confidence interval of the indirect effect contains 0 to determine whether there is an intermediary effect. The results of the Bootstrap analysis are shown in Table 7.

Table 6.
Results for the mediating effects of absorptive capacity

Variables	Model 9	*AC* (mediator) Model 10	Model 11	*PROD* Model 12	Model 13	*PROC* Model 14	Model 15
MARCON	−0.927***	−0.881***	−0.894***	−0.105	−0.056	−0.609***	−0.729***
Industry	0.408***	0.388***	0.389***	0.028	−0.008	−0.132	−0.194
Firm size	0.702***	0.807***	0.698***	0.254*	0.246*	0.251*	0.150
Firm age	−0.009	−0.011	−0.010	−0.009	−0.010	−0.001	0.002
LS	1.413***		0.063***		−0.013		0.221
BSS		0.275***	0.059***		0.338***		0.634***
AC				1.955***	1.879***	1.436***	1.346***
No. of obs.	1,088	1,088	1,088	1,088	1,088	1,088	1,085
Pseudo R^2	0.116	0.081	0.126	0.165	0.191	0.098	0.180
Log likelihood	−665.34	−691.60	−657.98	−629.23	−609.32	−605.67	−550.22
χ^2 (*f*)	175.13***(5)	122.60***(5)	189.84***(6)	248.50***(5)	288.32***(7)	132.15***(5)	240.84***(5)

Note(s): $*p < 0.10$; $***p < 0.01$

For the local search in Model 13, the interval (LLCI = 0.100; ULCI = 0.166) of indirect effect does not include 0, $p < 0.01$, whereas the interval (LLCI = −0.033; ULCI = 0.106) of direct effect contains 0. This demonstrates that a mediating effect exists, but no direct effect exists between local search and product innovation. In other words, AC plays an absolute mediation role between local search and product innovation. In terms of boundary-spanning search, the results verify a partial mediating effect of AC on product innovation. Moreover, the outcomes of Model 15 reveal the partial mediation role between local and boundary-spanning search and process innovation.

The moderating role of IT capability

Four hypotheses in this study proposed the moderating role of IT capability (including internal IT capability and external IT capability) in the relationship between knowledge search and innovation performance. We adopt a hierarchical regression to analyze the moderating roles with interaction terms created by centralized variables, and the results are shown in Table 8. In brief, we find a significant effect of IT capability on the relationship between knowledge search and innovation performance.

For product innovation, Model 16 shows that the coefficient of the interaction between local search and internal IT capability is positive and significant ($\beta = 0.433$, $p < 0.01$), supporting H6a. Meanwhile, Model 17 does demonstrate that external IT capability negatively moderates ($\beta = -0.144$, $p < 0.01$) the direct effect of boundary-spanning search on product innovation, implying that H6b is inversely supported. Furthermore, for process innovation in Model 18–19, the coefficients of the interaction between local search and internal IT capability and between boundary-spanning search and external IT capability are positively significant ($\beta = 0.788$, $p < 0.01$; $\beta = 0.127$, $p < 0.05$), which are in line with H7a and H7b, respectively. Thus, internal IT capability positively moderates the direct effect of local search on innovation performance. Nonetheless, in terms of external IT capability, it positively moderates the effect of bound-spanning search on process innovation but negatively moderates that on product innovation.

Based on the perspective of IT capabilities, the key to the success of knowledge search is that IT capabilities can help an enterprise to realize high efficiency of information sharing within or between organizations. This enables enterprises to strengthen the relationship between knowledge search and innovation performance with lower attention consumption. Specifically, internal IT capability plays an important role between local search and product and process innovation. This is mainly because internal IT capability can facilitate synchronization between the two. Internal IT capability can enhance internal R&D efficiency by shortening the timespan from quantitative change to qualitative change, and thus, promotes innovation. In addition, local search finds solutions from experience and practices,

Model	Variable	Effect	Coef.	S.E.	z	LLCI	ULCI
Model 13	*LS*	Indirect	0.133	0.017	7.88***	0.100	0.166
		Direct	0.036	0.035	1.04	−0.033	0.106
	BSS	Indirect	0.026	0.005	5.22***	0.016	0.036
		Direct	0.065	0.011	6.07***	0.044	0.087
Model 15	*LS*	Indirect	0.073	0.012	6.03***	0.049	0.097
		Direct	0.111	0.036	3.04***	0.039	0.182
	BSS	Indirect	0.009	0.003	3.00***	0.003	0.015
		Direct	0.095	0.008	11.75***	0.080	0.111

Note(s): Bootstrap = 5,000; ***$p < 0.01$

Table 7. Results of bootstrap mediation analysis

Variables	PROD Model 16	PROD Model 17	PROC Model 18	PROC Model 19
MARCON	−0.409**	−0.413**	−0.785***	−0.890***
Industry	0.136	0.182	−0.064	−0.089
Firm size	0.324**	0.417***	0.210	0.239
Firm age	−0.014	−0.014	−0.004	−0.002
LS	−0.626		−1.329***	
BSS		0.776***		0.166
IITC	0.221**		0.208**	
EITC		0.709***		0.573***
*IITC * LS*	0.433***		0.788***	
*EITC * BSS*		−0.144***		0.127*
No. of obs.	1,088	1,088	1,085	1,088
Pseudo R^2	0.113	0.136	0.174	0.211
Log likelihood	−668.38	−650.72	−554.03	−530.28
χ^2 (*f*)	170.21***(7)	205.52***(7)	233.22***(7)	282.92***(7)

Note(s): Bootstrap = 5,000; *p < 0.10; **p < 0.05; ***p < 0.01

Table 8.
Results for the moderating effects of IT capability

and the knowledge acquired by local search is consistent with the current knowledge base, which can reduce the possibility of cognitive dissonance.

However, the role of external IT capability varies by the type of innovation when companies practice a boundary-spanning search strategy. It strengthens the relationship between boundary-spanning search and process innovation but weakens the relationship between boundary-spanning search and product innovation. This may be attributed to types of knowledge brought by the boundary-spanning search. Boundary-spanning search can bring more heterogeneous knowledge for product innovation compared to process innovation. Employees are more likely to accept acquired knowledge that is consistent with the knowledge base. Although a large amount of heterogeneous knowledge can be provided by boundary-spanning search with the help of external IT capability for product innovation, a large difference will shock the employees' perceptions, which may lead to cognitive dissonance. Too much heterogeneous knowledge incurs employees' resistance. As a result, external IT capability weakens the relationship between boundary-spanning search and product innovation.

Discussion and conclusion

Main findings

Creating new knowledge by integrating internal and external resources into an existing organizational knowledge base is prevalent (Paruchuri and Awate, 2016). This research focuses on the relationship between knowledge search strategy and innovation performance, namely, product and process innovation. We explore the impact of different knowledge search behaviors on different types of innovation and probe the inner mechanism between them, including direct effect, mediating effect of absorptive capacity, and moderating effect of IT capability. The specific research conclusions are as follows.

First, in line with previous empirical research (Flor *et al.*, 2018; Ze *et al.*, 2018), our results show that the knowledge search strategies have a directly significant positive impact on product and process innovation. Furthermore, this study fills the research gap that few studies have examined the differences between the two search behaviors and two different innovative types. On the one hand, there are some differences when local or boundary-spanning search affect product and process innovation. Although local search affect product innovation and process innovation indifferently, boundary-spanning search has a significantly greater impact on process innovation than that on product innovation. On the other hand, the importance of local and boundary-spanning search is different in product or process innovation. For product innovation, the significantly positive impact of local search is greater than that of boundary-spanning search. Local search makes the enterprise more professional through the continuous accumulation of knowledge in a certain field so as to realize the development of new products based on existing knowledge assets. For process innovation, boundary-spanning search plays a vital role. The heterogeneous knowledge resources, obtained from outside, help a firm to increase the combination of different types of knowledge elements, which can provide the basis for process innovation.

Second, for the inner mechanism, this paper finds that the mediating role of AC is consistent with our expectations. Namely, the heterogeneous knowledge obtained by knowledge search needs to be transferred into the internal knowledge base through a firm's absorptive capacity. The vital role of AC between knowledge search and the innovative outcome has been proven by previous researches based on enterprise competence theory (Jeong *et al.*, 2019; Aliasghar *et al.*, 2019; Russo-Spena and Di Paola, 2019). This research merely provides complementary evidence from a large-scale sample of Chinese manufacturing firms. Although knowledge search may have a supporting role in innovation activities, firms also need the ability to embed searched knowledge in products or processes to improve performance (Ramayah *et al.*, 2020). AC not only reduces the costs of knowledge search by efficient identification of potentially valuable knowledge but also increases profits by better application of new knowledge to internal activities.

Third, we also test the moderating role of IT capability and find some interesting conclusions. The interaction of local search and internal IT capability has a positive impact on product and process innovation because internal IT capability can reduce innovation costs through integrating information search, knowledge storage, and communication with high efficiency. However, the interaction of boundary-spanning search and external IT capability positively affects process innovation but negatively impacts product innovation, which is partially contrary to our expectations. This can be explained by the heterogeneity of the knowledge brought by the boundary-spanning search.

Implications for research and practice

This paper contributes to the literature on knowledge search and innovation performance by building an integrative model that includes different types of firm abilities to explain how knowledge search behaviors work. First of all, our research clarifies the role of different knowledge search behaviors in promoting product and process innovation. And we further discover the difference between local search and boundary-spanning search on product and process innovation, which enriches the research results in the field of innovative search. Second, the result that absorptive capacity plays an absolute mediation role of local search and a partial mediation role of boundary-spanning search enriches the study of intermediary mechanism based on the enterprise competence theory.

Most importantly, this study reveals the mediating role of IT capabilities in different dimensions between knowledge search and product and process innovation. As an efficient approach to improving information processing speed, internal IT capability can enhance the ability of employees in acquiring, learning, and analyzing knowledge that can improve the understanding of the current knowledge base and future innovation directions. Meanwhile, external IT capability can promote communication efficiency with external patterns. On the one hand, the output of innovation mainly depends on the employees' subjective initiative. If employees are resistant to new knowledge because of cognitive dissonance (for example, NIH syndrome is the outcome of resisting external novelty knowledge), IT capability will not play its role well. On the other hand, the information system may reduce employees' personal contacts through which the tacit knowledge embodied in individuals and organizations can be transferred. This may be particularly true for Chinese firms that the relationship between firms or individuals always plays a critical role in communicating and knowledge sharing. To play the knowledge management role well, therefore, IT capability needs to be compatible with cognitive consistency. Enterprises should implement appropriate incentives to change attitudes towards new knowledge in order to better use IT capability to promote the impact of knowledge search behaviors on product and process innovation.

Moreover, our findings provide some managerial implications for practitioners on how to search and use new knowledge for innovative activities. First, fostering AC by implementing initiatives is a wise decision to fully benefit from knowledge search strategies for product and process innovation. The high level of AC encourages a firm to be more open in knowledge search (Wang and Guo, 2020), which states that R&D inputs are essential for innovation. In the second place, the results imply that IT capability acts as a vital contingent factor in searching for knowledge. Managers must take IT tools into consideration when implementing search strategies. IT improves the efficient sharing of explicit knowledge, but this advantage may vanish in regard to too heterogenous knowledge. Accordingly, managers should carefully formulate IT capability to intensify the positive side of knowledge search on product and process development. Last but not least, in the context of Chinese business culture, firms prefer a local search for product innovation but a boundary-spanning search for process innovation. Managers should balance these two behaviors to find optimal profits.

Limitations

Furthermore, a number of limitations should be addressed in the future. This study tested its hypotheses by using cross-section data from the World Bank. Some characteristics of the second-hand data impose certain restrictions on our study. Future research should tackle the relationships with a longitudinal study to validate this study's findings. In addition, the interactive mechanism between knowledge search and innovation performance may still be influenced by other factors that subsequent research can address. Moreover, the similarities and differences between China and other emerging countries, or between emerging countries and developed countries, may provide interesting results and shed insights on innovation at the national level.

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